

Genetic variations

5

5.1 Introduction

Human genetics is the study of the species that the majority of us regard as being the most significant on planet Earth: *Homo sapiens*. The desire to have a deeper understanding of ourselves is one motive for researching human genetics. The past 20 years have experienced robust human genetic variation. By understanding the genetic variation of an individual or a group, researchers are able to understand the organism and how they are able to adapt in a challenging environment. It was also observed that most of the variations are useless, which means the variation does not affect the ability to survive or adapt. Not all variations are positive or neutral; some of them have a negative effect on DNA sequence, which can cause diseases like Huntington's disease and cystic fibrosis. Finding out how genetic factors contribute to many human diseases is one of the advantages of studying genetic variation in humans.

The variations also contribute to molecular genetics, biology, and the biomedical field, which have exploded due to the number of variations in DNA sequence and genotypes that have been spawned. The human genetic data has increased the understanding of genetic variation patterns that occur in an individual or group, apart from the online database that is being provided in which genome variants are being published.

Homo sapiens hasn't had nearly as much time to build genetic diversity as most other species on the planet, which mostly predate mankind by vast epochs. The amount of genetic variation—biochemical individuality—between any two humans is approximately 0.1%. Any two (diploid) individuals have approximately 6×10 different bp, which is a significant factor in developing automated processes for analyzing genetic variation.

Most of the time, genetic variation is a natural thing that happens when DNA replication goes wrong. The term “variation” means a change in the genetic sequence. Due to genetic variations, all humans are different from each other in terms of hair, color, height, or even the structure of our faces. A group of individuals from the same species with similar characteristics have differences, these differences are due to variations. SNPs are the most prevalent form of variation in humans. The specialty of SNPs is that only one/single nucleotide is being replaced by the other

nucleotide in the whole DNA sequence. SNPs are dispersed naturally throughout the DNA of each individual. Approximately 4–5 million SNPs exist in a person's genome, since they appear around once every 1000 bases on average. A variant is deemed to be an SNP if it is present in approximately 1% of the population. Scientists have detected more than 600 million SNPs in global populations. The majority of SNPs do not impact development or health.

Nonetheless, a number of these genetic differences have been demonstrated to be crucial for human health research. SNPs assist in predicting a patient's reaction to certain treatments, vulnerability to environmental variables like pollution, and susceptibility to disease. It also helps find out how genetic changes linked to illness are passed down through families. These variations can be observed in genes, DNA, proteins, chromosomes, and even in the function of proteins. Although all the individuals have 99.9% similar genomes and only 0.1% of the variation in the genome, we are all different phenotypically. Thus, this small amount of variation accounts for such a significant difference. This difference may affect how an individual reacts to a particular drug or a particular environment. Thus, it is necessary to understand and evaluate the genotype-phenotype associations. Understanding such associations also helps us to devise therapies according to the patient's genetic makeup to provide a better therapeutic effect. Bioinformatics, or computational biology, has aided in pharmacogenomics and personalized medicine due to the integration of information available through whole genome sequencing and clinical studies. Several databases have contributed to the development of the field. These databases include PharmGKB, DrugBank, OMIM, and other such databases ([Lek et al., 2016](#)).

5.2 Types of variations

In humans, genetic variation can take many forms, some of which occur naturally and others which are influenced by environmental factors. Changes in single nucleotides, addition or deletion that causes a change in copy number, which ultimately changes the copy number variation (CNV), tandem repeats, recombination in homologous chromosomes, i.e., translocations and inversions (also known as copy neutral variations) are all examples of variation. The size of variation ranges from a single nucleotide to a megabase. Restriction fragment length polymorphism (RFLPs), single nucleotide polymorphism (SNPs), and indel are all types of single nucleotide substitutions or alterations in which a single nucleotide is changed or replaced from a specific locus in a DNA sequence ([Eichler, 2019](#)).

The chromosome microarray test is done to find out if there are any missing or extra chromosomes. Any piece of chromosome gained or lost is termed a “copy number variant.” All humans possess CNVs, most of which are harmless and do not cause any type of disorder. CNVs are a natural part of the evolution process. If the CNV in question is small and not linked to any important genes involved in the disorder, it is considered harmless. Also, CNV inherited from healthy parents is likely to be harmless. However, if the size of the CNV is large and contains an

important gene, it can be very harmful. But if the CNV is initiated in the offspring, then it can be very dangerous, and the person faces intellectual disabilities and multiple congenital syndromes (Fig. 5.1).

The next type of variation is the SNPs. Each SNP has four alleles. The SNPs are very abundant in number which can help in identifying the variation in the sequence. SNP is based on the oligonucleotide hybridization analysis which makes the detection of SNPs very fast and very precise.

Fig. 5.2 shows variations like single nucleotide changes, in which only one nucleotide is altered or replaced by another single nucleotide. GTAGGCCATGCA has been taken as the reference DNA sequence for all the variations. In the first case, single nucleotide substitution has been explained, in which the single nucleotide cytosine is being substituted by adenine. In this, insertion and deletion of single nucleotides are being observed. The second type of variation is tandem repeats; these repeats are short sequences of DNA that repeat themselves in whole. The variable number of tandems that repeat represents the total number of repeats in the whole sequence. The reason for insertion or deletion at a particular position is unknown.

5.3 Effects of genetic variation

The main effect of any kind of genetic variation is to give a population the flexibility to deal with environmental challenges. Previous experience has shown that genetic variation always helps and boosts the organism's chances of adapting to any type of challenging environment and prepares the organism for any type of unexpected situation. The organism reproduces to give birth to its offspring. The formation of offspring of the same kind is the primary reason for reproduction. However, it is not possible, to have identical offspring just like the parents, except in the case of clones, which are genetically identical to their parents. But in almost all cases, offspring are somewhat different from their parents. It is inevitable that there will be a difference between the parents and the offspring. Variations refer to the

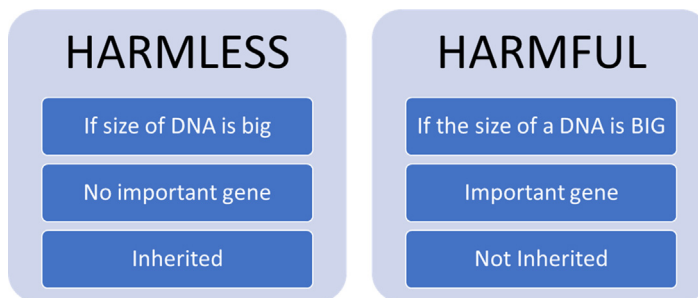
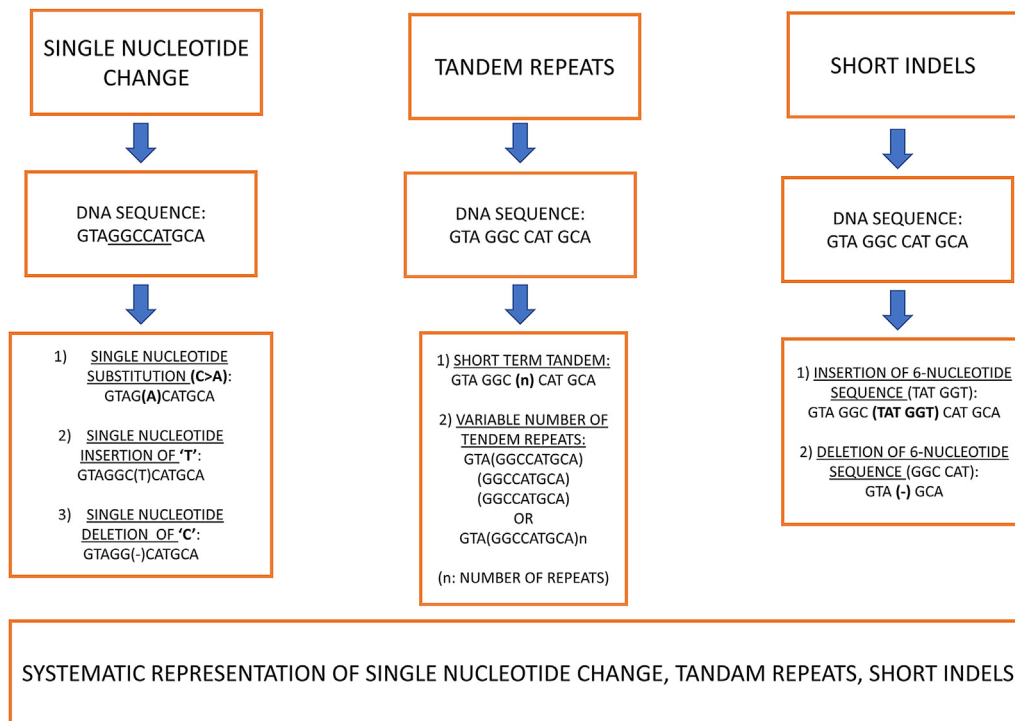


FIGURE 5.1

Characteristics of variations being harmful and harmless.

**FIGURE 5.2**

Systematic representation of single nucleotide change, tandem repeats, and short indels.

differences that exist between parents and their offspring. And these variations, over time, cause evolution. Evolution refers to any change in a population that takes years and improves organism capabilities. It is a very slow process and occurs in populations depending on environmental factors. The effect of any type of genetic variation is linked with evolution; the genetic variation is directly proportional to evolution. The more genetic variation, the more evolution; the less genetic variation, the lower the chances of evolution (Nei, 1983) (Fig. 5.3).

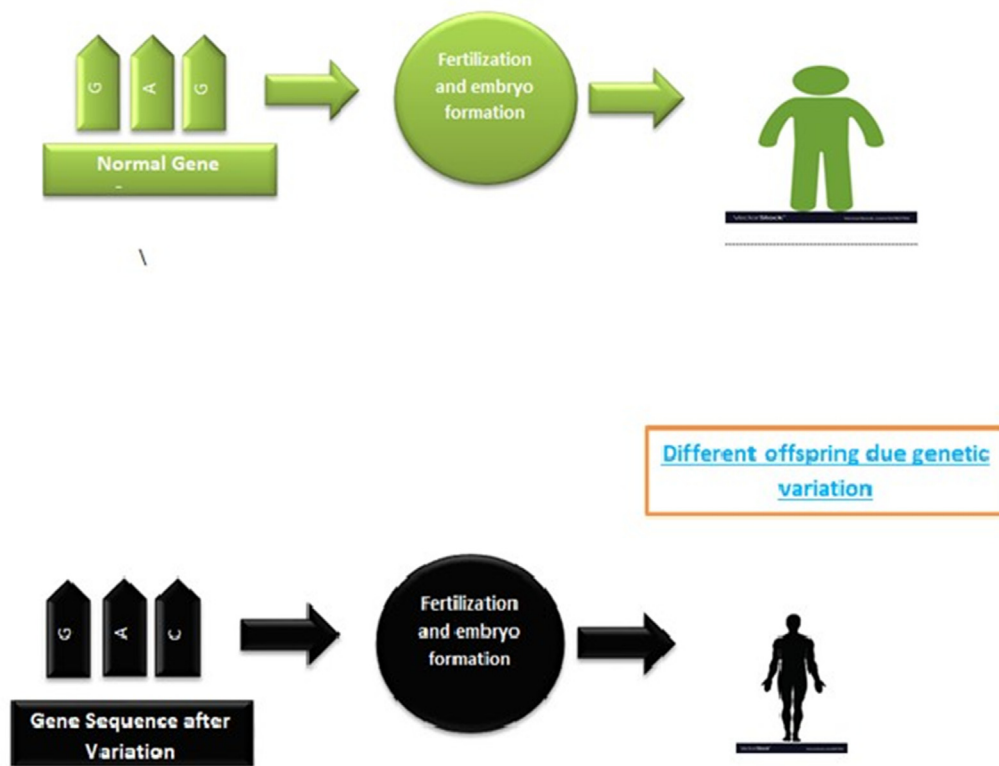
But still, not all genetic variation causes evolution; the variation that will be passed on to the next generation causes evolution in the population and is called hereditary variation. These variations are observed in the egg or sperm cell so as to pass on the trait to the next generation. The gene variant causes permanent changes in the sequence of DNA, which will further form the gene. In this type of genetic variation, it is not mandatory that it will cause a disorder; thus, the “gene variant” term is given to the permanent changes in the DNA. Gene variants are different from single nucleotide polymorphism because they can change more than one nucleotide. In single nucleotide polymorphism, only one nucleotide changes.

The variants can be passed from parents to offspring and occur over the course of an individual’s lifetime. If the gene variant is inherited, it is called a hereditary variant. The variation occurs in the germ cells, i.e., the egg or sperm cells of the parents, and is hence also considered a germline variation. When the egg and sperm cells containing variants form an embryo; the cells of this embryo will contain all the variant DNA from the parents and carry those variants through the rest of their lives. To function effectively, each cell requires hundreds of proteins to do their assigned jobs at the appropriate times and locations. Gene variations (also known as mutations) can occasionally compromise the function of one or more proteins. A variant can cause a protein to malfunction or not be formed at all by modifying the gene’s instructions for protein synthesis. When a variation affects vitally essential proteins, it may interfere with normal growth or produce a genetic condition, generally known as a disease caused by variations in one or more genes. Genes do not cause disease by themselves. Instead, genetic diseases are caused by changes that change or take away a gene’s function.

For instance, when individuals refer to “the gene of cystic fibrosis” they are often referring to a mutation of the CFTR gene that causes the disease. Every individual, including those without cystic fibrosis, has a variant of the CFTR gene (Studer et al., 2013).

5.4 Biological database

For the advancement of humanity, biology now incorporates numerous technological aspects. This expansion of biology has generated numerous data-rich outcomes that require compilation and organization. Databases are significant tools for scientists to use in analyzing and explaining a wide range of biological phenomena, from the structure of biomolecules and their interactions to the entire metabolism of

**FIGURE 5.3**

Different offspring's due to genetic variation.

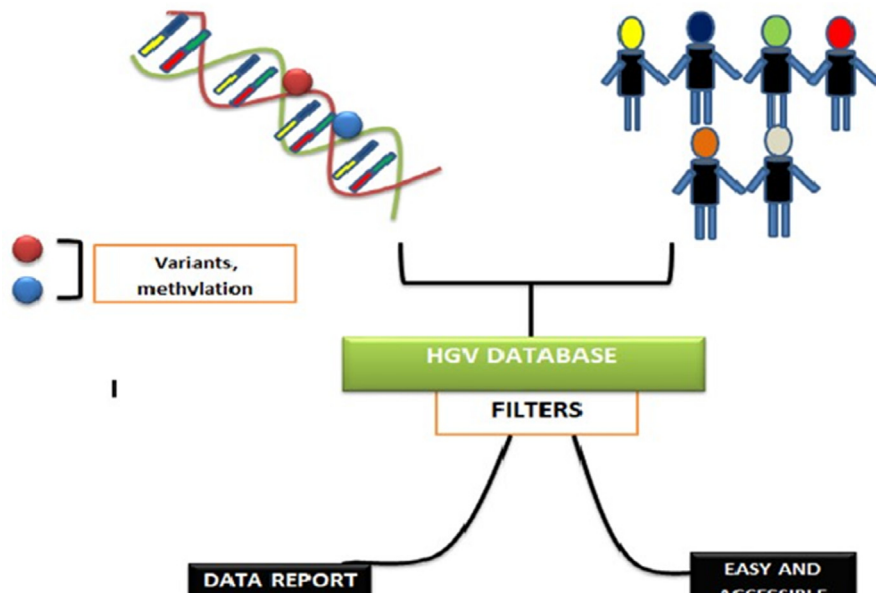
organisms and comprehending species evolution. This knowledge aids in the fight against diseases, the development of drugs, the prediction of specific hereditary diseases, and the discovery of fundamental links between species throughout the history of life. A database of genetic variants offers information about genetic variances. Researchers submit data to these databases, which collect, organize, and publicly document evidence linking a human genetic variant to a disease or condition. Based on the current level of knowledge, these databases may make statements regarding genetic variations. Understanding the relationships between genotypes (an organism's genetic code) and diseases or disorders can help in the diagnosis and treatment of people with genetic problems. A few of the genetic variant's databases are explained below:

5.4.1 Database of human genetic variation

There are various types of biological databases that help in collecting the data of various experiment results for further purposes. One of the biological databases is the Human Genome Variation Database (HGV) (Fredman et al., 2002). Human Genome Variation maintains a publicly available online collection of genetic variation as documented in published research reports. Following evaluation by the editors, a standard range of data about every variation is retrieved from journal-published articles and added to the library. It's a free database that may be accessed and utilized by anyone.

The search bar provides filters like region, mutation type, and zygosity type, which makes the search more accessible and easier. The HGVD website allows users to view the number of samples, genotype and allele frequencies, coverages, and eQTL (expression QTL) significances. The key point of HGV database is that it provides data reports which have all the short reports regarding the human genome variation. It also has reports regarding the variability, which has the main role of describing the disease causing "genomic variants" and also specifies the frequency of those variants. The data report and HGV database are interlinked with each other. Every piece of information in the HGV database is organized into "sets" that are connected to their corresponding data report, which will have a connection to the corresponding information in the database. The HGV database collects different types of data, involving OMIM data, de novo/inherited, name of the gene, chromosome, protein alteration, mutation type, GenBank accession number, disease, phenotype, region, codon base change, genome position, and references (Li et al., 2012) (Fig. 5.4).

ClinVar is a publicly available database of research detailing the associations between phenotypes and human variants, along with supporting documentation. It provides access to and facilitates discussion regarding the stated links between human variation and recorded health status, as well as the background of this assessment. It also handles submissions describing variations detected in patients' specimens, clinical significance claims, submitter information, and any supporting data. As per the HGVS guideline, the alleles mentioned in submissions are mapped to reference

**FIGURE 5.4**

Workflow of HGV database.

sequences and presented. ClinVar then represents the information for interactive users and those who want to employ it for various purposes and applications. It collaborates with interested groups to address the genomics community's requirements as effectively and efficiently as feasible. ClinVar is intended to facilitate the advancement of our understanding of the connection between genotypes and clinically significant phenotypes. By putting together information about variants found in people with and without phenotypes, the database helps prove the clinical validity of human variation ([Landrum et al., 2016](#)).

dbSNP: NCBI and NHGRI (National Human Genome Research Institute) established and operate dbSNP, a publicly available resource documenting genetic variation within and between species. It comprises a variety of molecular variation, including SNPs, small insertion and deletion polymorphisms (indels/DIPs), short tandem repeats (STRs) or microsatellite markers, and heterozygous sequences ([Sherry et al., 2001](#)). Since its establishment in September 1998), dbSNP has functioned as a primary, publicly accessible database for genetic variation. Once these variants have been detected and cataloged in the database, additional laboratories will be able to access the sequencing data surrounding polymorphism and experimental settings in future research applications. dbSNP presently classifies nucleotide sequence variants according to the following categories and percentages:

- I. single nucleotide substitutions, 99.77%;
- II. short insertion/deletion polymorphisms, 0.21%;

- III. invariant sequence sections, 0.02%;
- IV. microsatellite repeats, 0.001%;
- V. identified variants, 0.001%; and
- VI. uncharacterized heterozygous tests, 0.001%.

The 1000 Genomes Project It was initiated in 2008 and seeks to create the most comprehensive atlas of human genetic diversity by sequencing around 2500 genomes from approximately 25 groups worldwide. The genetic variation data provided by this multinational partnership will aid genome-wide association studies of complex variables and phenotypes, such as the pharmacogenomic phenotype. The purpose of a GWAS is to identify genomic areas related to an outcome of interest (e.g., drug response). Frequently, the related variation acts as a marker (tag SNP) for the true causative variant (the variant that underlies the observed association). To identify the causative variant(s) and understand the mechanism behind its effects, more analyses of variations within the candidate area are necessary. In several instances, uncommon variations within the candidate locus contribute to disease susceptibility or phenotype. But these rare changes aren't in the HapMap database and aren't picked up by existing genotyping technologies, so they aren't taken into account in GWASs (Li et al., 2012).

5.4.2 Predicting the clinical significance of human genetic variation

Any type of testing, diagnosis, or risk assessment of the patient and their family for any type of genetic disease is called molecular genetic testing. Today, there has been a robust development in molecular genetic testing that is low-cost, fast, and efficient. The DNA sequencing technology and computational technologies combined increase the standard of molecular genetic testing, which shows the result that there is more DNA variants per test due to lifestyle and environmental factors, which may include pollution and global warming. To determine the pathogenicity of the DNA variants, various approaches and methods have been organized in a systemic way that can help overcome this increased number of DNA variants. Initially, the database provides existing data, which includes publication of internal and public resources. After that, the statistical analysis is being performed, which includes the assessment of population and disease. Now, both the in vivo and in vitro experiment data is being evaluated on the basis of computational prediction, which basically predicts the impact of each variant. Finally, with all the evidence, we can predict whether the variants will cause the disease or not (Duzkale et al., 2013).

A report suggests that the most common result of molecular genetic testing includes DNA variants and cystic fibrosis, or a gene that is linked with the disease and has a sequenced coding region. Next-generation technologies have improved diagnostic sensitivity while also providing high throughput results at a low cost. Such technology includes next-generation sequencing (NGS). It is also observed that the variants with uncertain significance (VUS) have also increased, which

basically means that the genetic sequence for a particular disease is not clear. The functionality of the variant identified by the genetic testing is unclear. Identifying variants also helps us figure out what role a gene plays in a certain disease. Genome sequencing, on the other hand, adds layers of complexity to the questioning of the gene.

The molecular genetic testing has had great success in the diagnosis and evaluation of genetic diseases, providing high throughput results at a low cost. However, protocols, rules, regulations, and ethics are lacking in molecular genetic testing. When it comes to variant interpretation, there are some guidelines available that are updated as variant knowledge expands, including the American College of Medical Genetics and Genomics (ACMG). There are three questions that can be used to figure out what the clinical significance of a variant is for a certain disease.

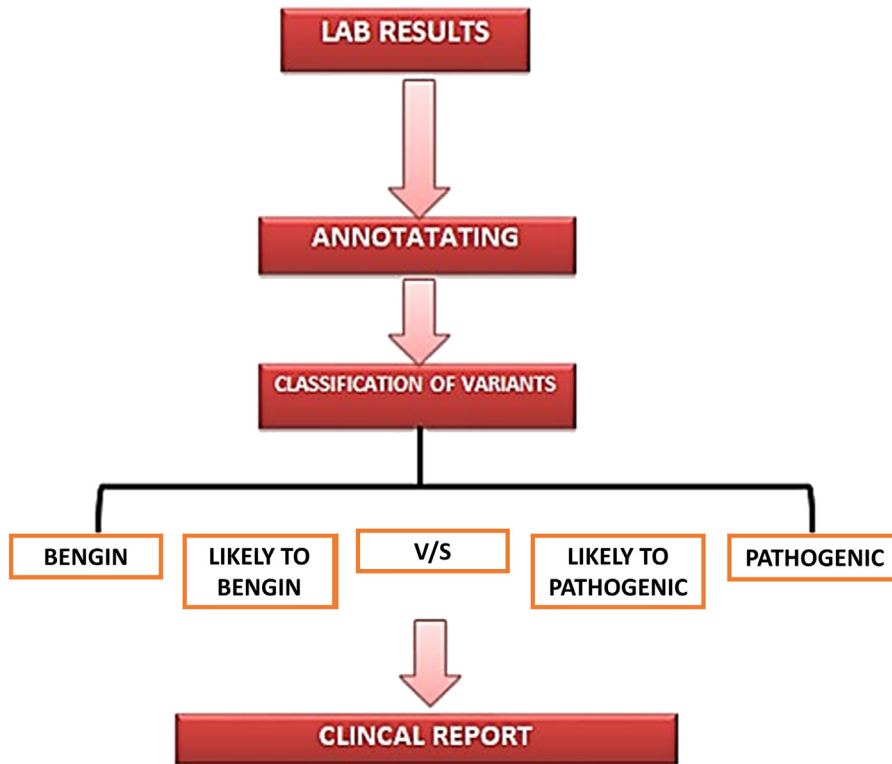
- a. Does genetic variation also change the functionality of a particular disease?
- b. Is this variation cause disease?
- c. Is there any relation between the associated disease and the present clinical condition of the patient?

While the interpretation of variant questions 1 and 2 is very important to assess the variant and its effect, But in some cases, to maximize the benefit to the patient, one should do the entire set of three questions, which provide a three-dimensional interpretation of the variant. This is especially true in the case of variants with uncertain significance (VUS). While testing 21,000–22,000 cases in a case study, it was discovered that 246 genes are linked to 50–56 human diseases. For clinical assessment, a systematic framework has been made, and more than 17,000 variants are being found and about 8000 of them are being recognized in patients.

The Fig. 5.5 shows the systematic representation of variant assessment. The genetic variation is acknowledged by the test which is performed by the labs. After that the results are being annotated on the basis of publication, internal database and computational prediction algorithm. The result which are got from annotation in the form of clinical data are being classified by the expert individual into five sections, benign, likely to benign, variants with uncertain significance (VUS), pathogenic, likely to pathogenic (McLaren et al., 2016).

5.5 Phenotype-genotype association

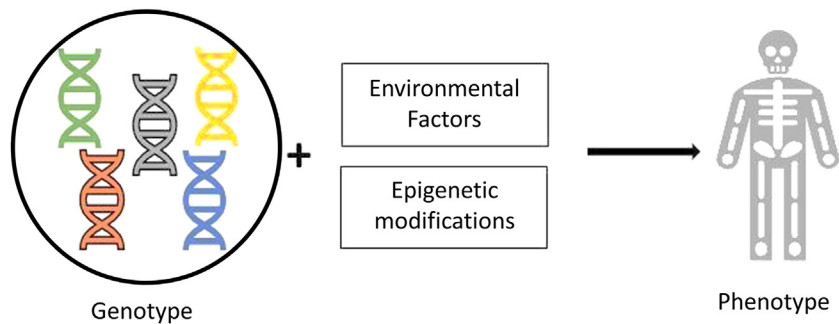
Genotype refers to the entire collection of genes present in an individual. The genotype is inherited from both parents and thus governs heredity and the extent of development. Sexually reproducing organisms have unique genotypes due to the fusion of gametes and the process of crossing over during meiosis. The characteristics of an individual are the phenotype and are governed by the genotype and other factors. Observable features or physical attributes of an individual or organism are referred to as their phenotype. Individual phenotypes are the result of a complex interaction between an individual's genotype and other factors such as epigenetic modifications

**FIGURE 5.5**

A typical variant assessment workflow.

and environmental factors. The phenotype includes color, height, shape, behavioral characteristics, biochemical processes, and many more. The phenotype is subjected to constant change due to ever-changing environmental conditions and the morphological and physiological changes associated with aging. The environmental factor influences the expression of the genotype. The association between genotype and phenotype is complex because, despite having 99.9% similar genomes, all individuals differ phenotypically to a great extent; Fig. 5.6 depicts the complex genotype-phenotype association.

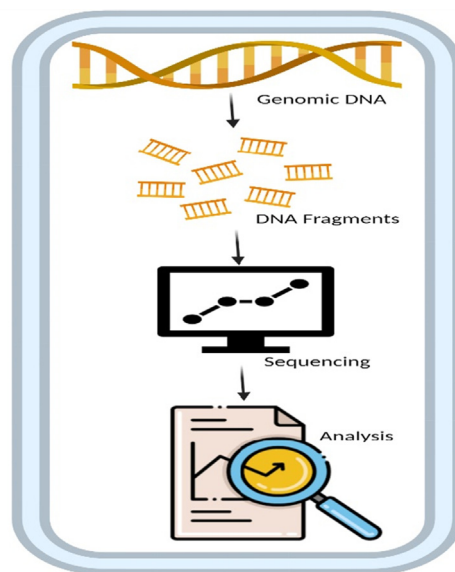
The phenotype of an organism can be observed through simple, direct observation, whereas genotyping is done to observe or characterize the genotype. Genotyping involves various complex procedures, from the extraction of DNA to sequence analysis. Previously, genotyping of partial sequences was the only option, but with technological advancements, whole-genome sequencing can now be done quickly. Partial sequencing does not provide much information because most of the proteins are encoded by widely spread gene fragments, which may be present at different loci too. WGS makes it possible to identify the full sequence of a person, which can then

**FIGURE 5.6**

Schematic representation of the complex relationship between genotype and phenotype.

be analyzed and compared to a phenotypic trait (usually to find the cause of a disease). Fig. 5.7 shows a schematic representation of the WGS workflow. The other methods used for genotyping include DNA microarrays, PCR, DNA hybridization, and others.

The understanding of genotype-phenotype association is a developing field of research, especially in the context of pharmacogenomics. Individual differences determine how an individual will react to a drug or how susceptible he or she is to contracting a disease. For example, variations in the *CYC450A* liver enzyme

**FIGURE 5.7**

Schematic representation of the WGS workflow.

involved in the metabolism of several medicines, can influence a person's capacity for metabolizing a certain drug. This information about genetic differences can help doctors figure out the right amount of the drug to give for effective treatment.

Integrating both genotyping and phenotyping techniques is more adequate for a diagnosis or association study than genotype tests alone. The integrated approach is used in pharmacogenomics to identify the disease cause and drug metabolism with greater accuracy than predicted using genotyping alone. This integration also becomes the baseline for the personalized medicine approach. The genotype-phenotype association can easily be studied using knockout mice.

The field of genomics underwent a drastic transition with elucidating the human genome sequence and mapping of the SNPs (single nucleotide polymorphism). With technological advancement, it is now possible to analyze hundreds of SNPs at the same time. This feasibility in analysis has allowed the genotype-phenotype association to establish reproducibly. For example, SNPs of *TCF7L2* (transcription factor) and *PPARG* (peroxisome proliferator-activated receptor- γ) are associated with diabetes. Similarly, *NOD2* (nucleotide-binding oligomerization domain containing 2) SNP is associated to Crohn's disease, and *CFH* (complement factor H) SNP is associated with age-related macular degradation. Genotype-phenotype association studies are not only relevant for pharmacogenomics but are also essential to study the evolution of a particular trait. These studies help in the elucidation of genes responsible for behavioral traits. For most phenotypes, genes responsible are spread over different parts of the genome, which shape the phenotype differently under different environmental conditions. Due to this variable spread of genes encoding a phenotype, genome-wide associations are limited for such phenotypic traits.

Tremendous efforts have been made to link the human genome variations with the phenotype, and this process has accelerated over the past few decades. These studies have disclosed various phenotypes/traits affected by the variation present in the genome. To interpret the phenotypic prediction based on variation, a thorough understanding of genotype-phenotype association is required.

The various methods used for detection of genetic variation include the Mendelian approach of crossing and assessing the phenotype, but this method is not possible with humans due to the longer generation time. Other approaches include PCR based methods, FISH, RFLP, sequence analysis, and microarrays. These newer techniques are fast and precise, allowing multiple detections in a single pass.

5.6 Pharmacogenomics

Pharmacogenomics/pharmacogenetics refers to the study of genetic variations that cause variability in how different people react to drugs. In 1994, a study published in the USA presented the statistics that around 2.2 million people were severely affected by the drug side-effects, and around 100,000 people died due to the side-effects. According to these statistics, the death rate from drug side effects was higher than the death rate from viral diseases. Predicting how a patient will react to a

specific drug will thus be a significant advance in the health sciences (Relling and Evans, 2015).

The individual's response to a particular drug depends on the target protein or enzyme and the receptors responsible for binding the drug and metabolizing it. Variations in such proteins at the genetic level may result in decreased binding or even the complete absence of drug binding to receptors and an absence of or decreased drug metabolism. The genetic composition of an individual can affect how the body reacts to the drug. Some factors that may be affected due to the genetic variation include:

5.6.1 Drug receptors

Certain drugs are attached to a particular target receptor to function effectively. The genetic composition determines the type, frequency, and specificity of the receptor, which may affect the way your body reacts to a particular drug. Depending upon the genotype an individual possesses, a higher or lower amount of drug may be needed, or a different drug may also be needed, as depicted in Fig. 5.8. Example: In the case of some breast cancers, *HER2* receptors are overexpressed, which helps in the development and progression of cancer. The *T-DM1* drug targets the *HER2* receptor of the cancerous cells and eventually kills them. If a person has breast cancer, then the clinician tests the tumor sample to check whether the samples are *HER2* positive or negative and then decides whether *T-DM1* can be used for treatment or not.

5.6.2 Drug uptake

Certain drugs need active uptake inside the target tissues or cells for their practical function. The genome of an individual affects the uptake of certain drugs. Decreased permeability toward a drug hinders the cellular uptake of the drug, leading to the accumulation of the drug at random sites, which may cause side effects. The genetic makeup of a person affects how efficiently a drug can be taken up by the target cells and how long it takes to be excreted out of the cells. If the drug is excreted too fast, it might not create the desired therapeutic effect, as shown in Fig. 5.9. Example: Statins are a class of drugs that help lower the cholesterol level by acting on the liver. For statin activity, they must be transported to the liver. This transportation is facilitated by a protein encoded by the gene *SLCO1B1*. A variation in the *SLCO1B1* gene causes less uptake of a statin known as simvastatin into the liver. If administered at a higher dosage, simvastatin accumulates in the blood, which leads to problems in the muscles, which include pain and weakness. The clinician usually recommends the genetic testing of *SLCO1B1* before prescribing simvastatin.

5.6.3 Drug breakdown

The genomic construction of an individual may affect the rate at which the drug is metabolized. If a drug gets metabolized faster than expected, it is excreted from the

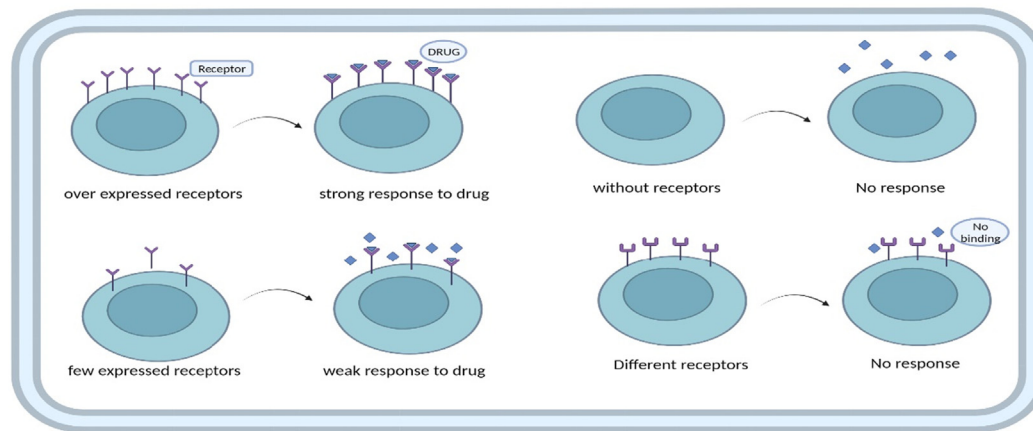
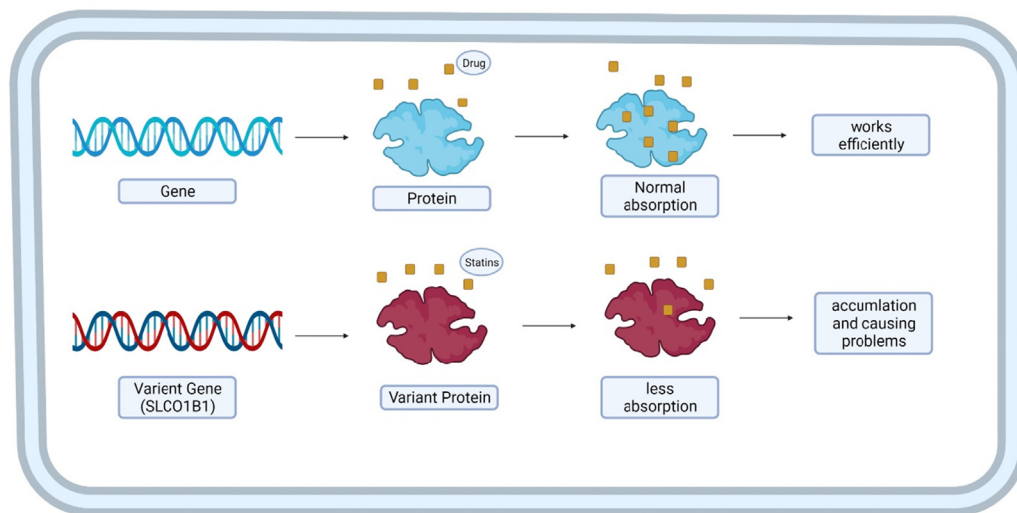


FIGURE 5.8

Variation in response to a particular drug due to genetic variations amongst individuals.

**FIGURE 5.9**

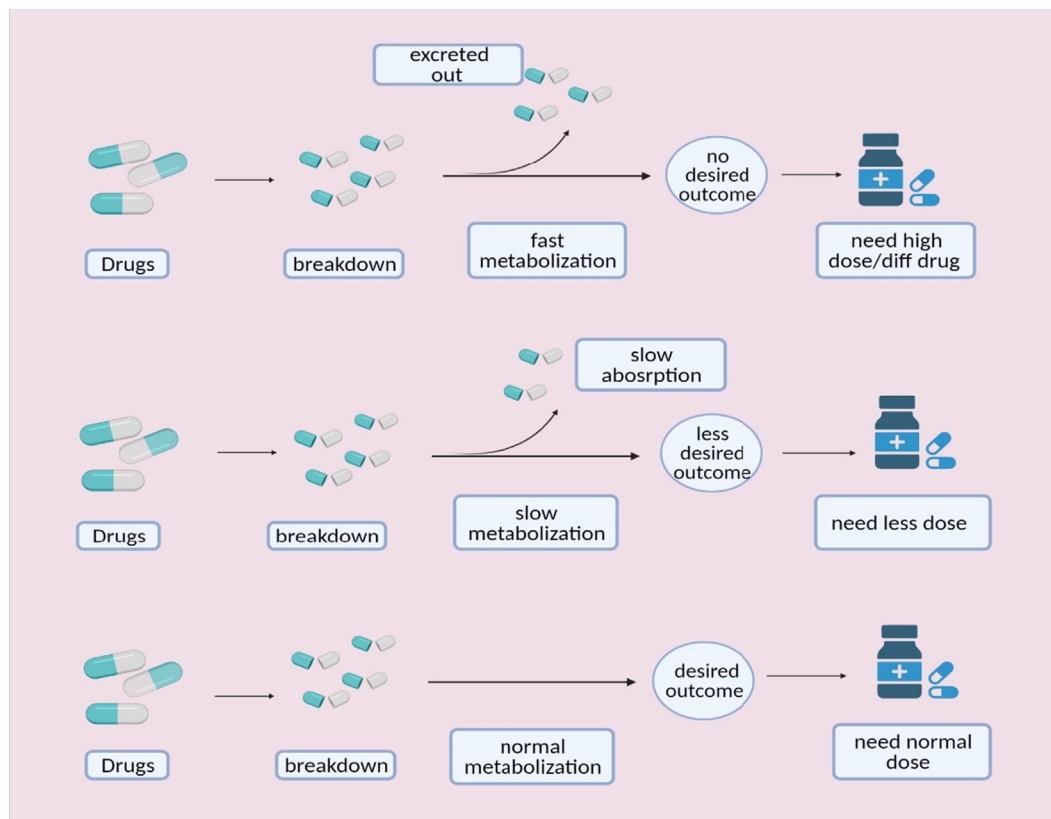
Variability in the drug uptake caused due to genetic variation.

body in a shorter period of time without causing the desired effect. In these cases, the drug dose needs to be increased or a different drug needs to be used. If the body metabolizes a drug more slowly than expected, a lower dosage is needed, as shown in Fig. 5.10. Example: The breakdown of amitriptyline, an antidepressant drug, is controlled by two genes named CYP2D6 and CYP2C19. A psychiatrist, before prescribing amitriptyline, generally recommends a genetic test for both genes. This testing helps identify the optimum dosage for the drug. If the genes are highly active, then the drug will breakdown faster, and either a high dosage will be prescribed, or a different drug will be prescribed. If the drug breakdown is slow, then a lower dosage will be prescribed.

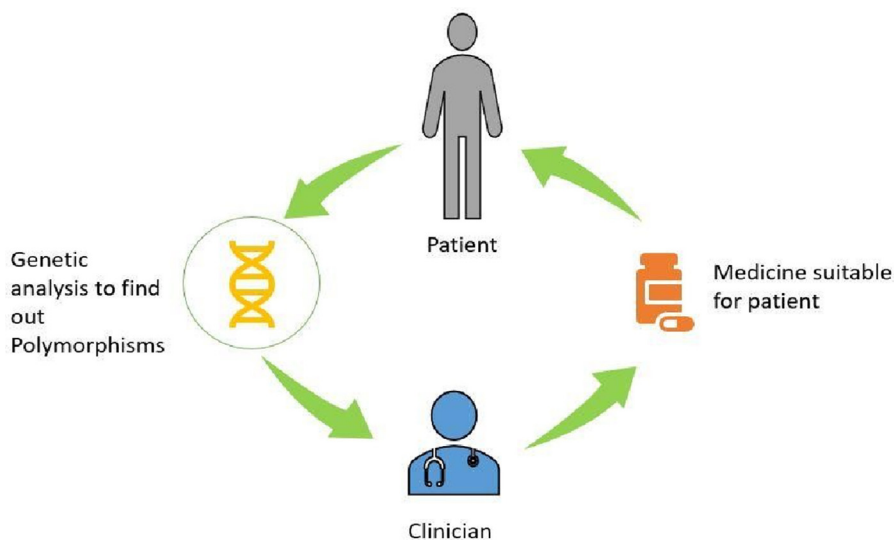
One of the essential polymorphisms is found in the protein family, cytochrome P450. The enzyme CYP2D6 is essential for the breakdown/metabolism of about 20%–25% of all the drugs prescribed by physicians. Mutations in the gene encoding CYP2D6 can affect the rate of drug metabolism. Based on the type of mutation, the patients are classified as ultrafast drug metabolizers, extensive drug metabolizers, medium drug metabolizers, and slow drug metabolizers. This suggests that genetic polymorphism or genetic variation influences a patient's ability to respond to drugs. SNPs, being the most widely and frequently found genetic variation, are at the center of pharmacogenomics, as scientists are focused on determining the SNPs that alter the drug metabolism in an individual (Mateo et al., 2022).

Pharmacogenomics aims at predicting the unwanted side-effects of drugs used for therapy in advance. For predicting in advance, diagnostic test development is one of the prerequisites. This diagnostic test helps the clinician find out if any genetic predispositions are present in the patient and how the patient may react to a particular drug. Based on the diagnostic tests, the clinician or the geneticist finds out whether a distinct polymorphism is present in the drug-metabolizing enzymes or not. Based on the test results, the patients are then classified into various groups, and a suitable therapy is administered to them based on their genotype, as shown in Fig. 5.11. This type of therapy is also known as “stratified medicine” because the therapy is molded or tailored according to patients' genetic makeup.

Example of “Stratified Medicine”: Mercaptopurine and thioguanine are used as chemotherapeutic agents to treat acute lymphatic leukemia (ALL). These drugs incorporate into the DNA of cancerous cells, eventually killing them. Thiopurine-S- methyltransferase is the enzyme responsible for the metabolism of these drugs. Genetic polymorphisms may interfere with the activity of this enzyme, leading to the accumulation of these drugs in blood cells and causing death. In contrast, in patients with high enzyme activity, high concentrations of drugs are needed. Pharmacogenomics also aids in the process of pharmacological research. Before entering the market, every drug must undergo rigorous clinical trials to prove its safety and efficacy. Pharmacogenomics helps exclude the cohort that may show no effect of the drug or may show unapprehensive side-effects before the start of the clinical trials. This filtering out of the cohort may increase the probability of the drug reaching the market. Pharmacogenomics also helps in the development of drugs for patients who do not respond to any of the therapies that already exist.

**FIGURE 5.10**

Schematic representation of different rates of drug breakdown.

**FIGURE 5.11**

Schematic representation of finding genetic predispositions of the patient and tailoring drugs that would be more effective for treatment.

5.7 Pharmacogenomics and targeted drug development

Drug development using the pharmacogenomics approach focuses on treating the cause of the disease instead of just treating its symptoms. Some of the clinical diseases are caused by mutations in genes. Different types of mutations can be found in the same gene, which may have altogether different effects. Some mutations may cause a protein to lose its function, while other mutations may not lead to protein synthesis at all. Drugs can be designed based on the types of mutations that affect the protein. These drugs will be effective only against a specific type of mutation. Example: Ivacaftor is a drug used for the treatment of cystic fibrosis. A mutation in the CFTR gene leads to an alteration in the CFTR protein responsible for forming ion channels. The mutation causes this channel to be permanently closed, thus preventing the movement of particles. Ivacaftor forces the ion channel to open again, but this drug will be ineffective if the ion channels are not formed at all ([Whirl-Carrillo et al., 2012](#)).

Despite the genetic variation, other factors affecting the response and efficacy toward a drug include the patient's age, alcohol consumption, nutrition, the individual's microbiome, the presence of other diseases, and the consumption of other drugs. However, the presence of genetic variation can, but not necessarily will, lead to variation in metabolic pathways. So, to increase the efficiency of individual medicines, genetic predispositions and metabolic profiles should be considered. Thus, an integrated approach that incorporates both pharmacogenomics and pharmaco-metabolomics results in more tailored medicine.

5.7.1 Personalized medicine

Personalized medicine (PM) is a relatively new discipline of medicine that examines a person's genotype to aid in disease prevention, treatment, diagnosis, and cure. The genotype of a patient can be utilized to determine the optimal medicine or treatment option for them. The Human Genome Project's data is being used to enhance personalized treatment. Personalized medicine is a planned, effective, and preventative structural method for effective health care. To account for inherited predispositions, PM engages individuals in lifestyle decisions and effective management of health. PM is now possible and growing at a quicker rate thanks to significant technological advancements, including as: New methods for sequencing the human genome faster and more correctly with smaller but more powerful equipment ([Whirl-Carrillo et al., 2012](#)). Large-scale research and sample archives that link genetic variants to disease across countries and continents ([Fig. 5.12](#)).

5.7.2 Personalized medicine drivers

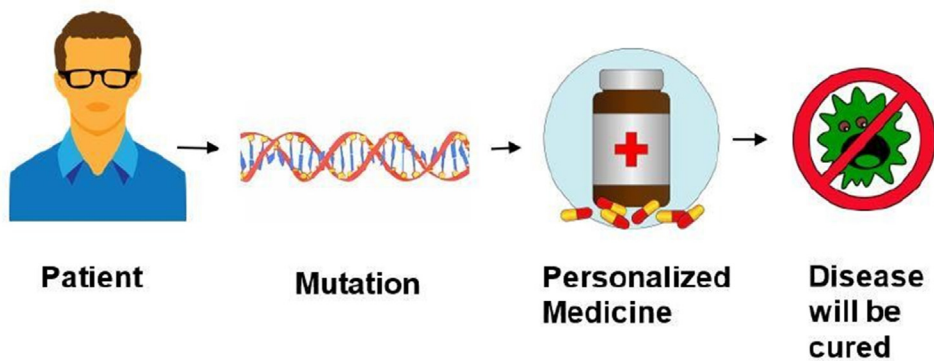
5.7.2.1 *Human genome sequencing has been completed*

The rise of information and knowledge outlining the causes of disease and patient heterogeneity, as well as heterogeneity in treatment response, will be the primary driver for customized medicine. The goal will always be to enhance the benefit-to-risk ratio associated with the administration of a particular medicine to a specific patient, as well as to identify newer, more specific, and safer medicines.

The emerging knowledge of medical science reveals that the heterogeneity of diseases and patients will almost certainly assist in the discovery of new disease-related objectives and more sensitive and accurate diagnostic tools. A standard, complex, chronic disease is thought to be caused by as many as 10 distinct genes on average. This should lead to a variety of new and improved therapeutic approaches as well as earlier interventions. As a result, healthcare practitioners will have a wide range of pharmacological options to treat patients with various diseases. When diagnostics packaged with pharmaceuticals become accessible, they will help health care providers decide which drug option is best for each patient. The health care provider will subsequently prescribe correctly. This decision will be based on drug efficacy, safety, or a combination of the two. Although no one knows when this will be possible, many, if not all, diseases will undoubtedly be several years away.

5.7.2.2 *Molecular characterization of disease*

As more information on the disease and patient heterogeneity becomes available, relevant genomic, genetic, and proteomic data linked to clinical information in well-characterized people will be collected and thoroughly mined utilizing bioinformatics and statistical methods. High-throughput genotyping, as well as gene and protein expression approaches, will be used to achieve this. This is already being done in several diseases, including cancer. It identifies somatic changes in gene expression and links them to interpatient variability in treatment response, as well as selecting appropriate therapeutic strategies. As a result of the timing of the

**FIGURE 5.12**

Schematic representation of the workflow of Personalized Medicine.

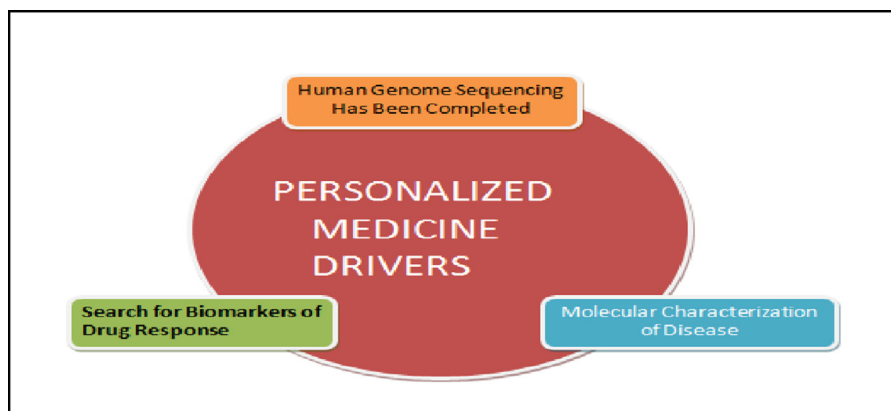
intervention, this could lead to a plethora of targets, and, eventually, medicines that will provide the most significant benefit to the illness classification will help to stratify at-risk patient populations further, resulting in early disease detection or prevention, more targeted therapeutics, and markers that can help guide drug therapy decisions. In some disorders, it may soon be possible to identify the genes and targets that have a significant role in disease development or penetrance.

Identifying all genes and targets involved in disease penetrance will take several years. However, using SNP maps, it is now possible to compare genetic differences between drug responders and non-responders. This could have an impact on how clinical trials are designed and on drug development for specific disorders. Similarly, while there is an increasing desire for safer pharmaceuticals, particularly in the treatment of chronic disease, there will be an increasing demand to use drugs in those who are most likely to benefit from them rather than those who will simply experience adverse effects. All of this will require the development of a number of markers and safety correlations that will help find the molecular cause of disease in a specific person.

5.7.2.3 Search for biomarkers of drug response

The role of environment, the level and complexity of the disease at a given time, the influence of drug-drug interactions, the person's overall health, including primary organ function, and disease complications are the parameters that lead to heterogeneity. In 1962, the first comprehensive treatment of pharmacogenetics was published. Because even the most acceptable medicines do not lead to substantial efficacy or safety in 100% of treated individuals, one option for personalized medicine could be to increase efficacy, safety, or both for an approved treatment. SAEs have been responsible for the withdrawal of drug candidates during drug development, as well as the removal of authorized pharmaceuticals after launch, due to genetic variance in drug efficacy. The role of environment, the level and complexity of the disease at a given time, the influence of drug-drug interactions, the person's overall health, including primary organ function, and disease complications are the parameters that lead to heterogeneity. In 1962, the first comprehensive treatment of pharmacogenetics was published. Because even the most acceptable medicines do not lead to substantial efficacy or safety in 100% of treated individuals, one option for personalized medicine could be to increase efficacy, safety, or both for an approved treatment. SAEs have been responsible for the withdrawal of drug candidates during drug development, as well as the removal of authorized pharmaceuticals after launch, due to genetic variance in drug efficacy ([Fig. 5.13](#)).

Biomarker-associated treatments for a variety of infections are obtainable and may explain clinical variability, drug response risks, particular doses, dose approaches, and polymorphic drug targets. The FDA-approved drugs for certain diseases along with their biomarkers are mentioned in [Table 5.1](#). It describes medication that must be recommended to individuals with biomarker-related adverse effects. This can be extremely helpful in providing a safe and efficient

**FIGURE 5.13**

Factors contributing to the developments in the field of Personalized Medicines.

Table 5.1 Biomarkers of pharmacogenomics for the disease therapy.

Name of drug	Area of therapeutics	Biomarkers	Referenced subgroups	Efficacy
Afatinib (or tyrosine kinase inhibitor)	Oncology	EGFR	EGFR exon 19 deletion or exon 21 substitution (L858R) positive	Such changes bring sympathy in the treatment of afatinib
Carvedilol	Cardiology	CYP2D6	CYP2D6 poor metabolizers	Higher concentration of carvedilol plasma, dosage caution is needed
Celecoxib	Rheumatology	CYP2C9	CYP2D6 poor metabolizers	Lower dosage recommended
Diazepam	Psychiatry	CYP2C19	Poor metabolizers of CYP2C19	Lower dosage to rule out extended sedation
Abacavir	Infectious diseases (like HIV)	HLA-B	Allele carriers of HLAB5701	Higher risk of immune mediated hypersensitivity reactions

treatment based on biomarkers. The finding of novel and potential biomarkers associated with the infection may play a prominent role in the identification, placement, and behavior of the treatment itself.

5.7.3 Future aspects of pharmacogenomics in personalized medicine

Managing the severity of the condition is one of the most difficult aspects of treatment. In the following generation, a considerable number of novel mutations are observed. Although the relationship between some of these mutations and diseases is widely understood, the effect of multiple alterations/mutations has yet to be investigated. In disease forecasting, some changes may have individual or collective implications. Systematic retrieval, retrieval, and analysis are required to keep clinical data and make decisions. Better disease biomarkers can aid in diagnosis and treatment. Because a better diagnosis of the problem can improve the quality of treatment, there needs to be financial motivation to come up with new diagnoses.

Microbiome research is now possible thanks to advances in DNA sequencing. Microbiome research focuses on diseases and could lead to more personalized treatments. It is vital to identify the connection between genetic variation and its associated consequences.

Pharmacogenomics was born out of these organizations, and it no longer focuses just on clinical validation and application. The clinical utility and cost-effectiveness of pharmacogenetics tests should be considered. Data and trials relating to pharmacogenomics are only available for other medications, but this information isn't being turned into medical services on a large scale. However, the clinical use of pharmacogenomics testing is limited due to the disparity between its clinical performance and its use in individuals by health care professionals. Some drugs include pharmacogenomics-based CPIC dosing guidelines. Some strategic recommendations have been proposed because primary care practitioners lack adequate criteria to assure the success of tailored drugs. Some of the suggestions are to use more biomarkers, make a genetic code that protects privacy and anonymity, make sure drugs are available to everyone in a fair way, test and document genetic information, make sure genetic testing is regulated well, and make more people aware of drug-modified pharmaceuticals.

5.8 Computational biology methods for decision support in personalized medicine

Understanding studies of various levels and periods, like the incorporation of genetic information (i.e., genotypes) with a patient's data on medical history, remains an issue (i.e., phenotypes). Bioinformatics plays a very significant role in assisting integrative analyses at different levels to draw conclusions in support of more errorless

detection & its solutions, as well as creating strict patient recommendations with desirable precautionary therapeutic options. Such strategies would enable health providers to provide their patients with the “proper information at the right time, in the correct form, to the correct people.”

Data integration combines laboratory and clinical environments for translational bioinformatics methodologies, enabling more efficient management and cost reductions. To increase peculiar patient’s medicine in the medical field, information at many levels, like cells, genes, proteins, medications, and functional aspects, must be incorporated within the phenotypic data in EHRs. Also, data incorporation will lead to the development of large databases to generate new information and skill proficiency. E.g., a web-based database with data mining and inquiry tools known as the RNA-SEQ Atlas was created for gene expression profiling. Other practical databases, microarray profiles, signaling pathways, & genetic ontologies are all linked through the integrative system.

Such comprehensive methods can be beneficial for comparing tissue-specific expression profiles to uncover trends, as well as correlating tissue functions with genetic modifications to enable other decision-making processes at multiple system levels. Collecting various origins of data that meet the domain & necessary inspection is a crucial stage in data integration. Many of these records are disorganized and include errors, and they must be repaired, cleaned, organized, and updated. Both redundancies and inconsistencies need to be eliminated and addressed, respectively. One gene, for example, could have numerous entries with various names. To bring the shared values together, such concerns must be handled using GO (gene ontology). Other techniques, such as the CABIG (Cancer Biomedical Informatics Grid), aid in the resolution of such complex problems.

Standardization is a critical step in resolving interoperability issues in several medical and laboratory settings across a wide range of knowledge domains. For example, the SNOMED CT (Systematized Nomenclature of Medicine Clinical Terms) & ICD (International Classification of Diseases) is frequently used for clinical data and billing processes. Clinicians use the Digital Imaging and Communications in Medicine (DICOM) standards to process imaging data. Rx Norm Resource is a database of drug terminologies that connects them. KD (knowledge discovery), a critical element of decision support, can be performed through a reiterative and dependent process involving data integration and data mining technologies.

These approaches are required for finding successful pharmacological targets and employing personalized treatments. For systematic studies and vigorous assessments, like chronobiology research, data mining is essential. Data mining techniques can identify spatiotemporal patterns and develop grouping, association, and dependency models. Neural networks, based on probability graph models like Bayesian networks, text data mining, and evolutionary design are just a few of the data mining techniques available. Agent-based modeling may describe complex nonlinear systems at various different living dimensions, from biological cells to different societies. For instance, many knowledge discovery and computational biology mythologies, such as BLAST, MATLAB, and clustering algorithms, are adopted.

Gene expression in lung tissue can be measured by these methods by inspecting the patterns of the dark/light circadian cycle. Dynamic thermal analysis (DTA) was discovered to be very helpful for the recognition and identification of cancer (breast cancer) using artificial neural networks (ANN). For inspecting the etiology of breast cancer to designate the multifaceted essence of the disease, an agent-form archetype of mammary ductal epithelium dynamics was utilized. It depicts the multifaceted essence of various diseases, which includes both cellular and molecular causes. Furthermore, semantic Web technologies have been recommended as helping organize and display pharmacogenomics knowledge related to the generation of various drugs and decision-making in various medical fields. Such informatics technologies have the potential to become critical components of CDSSs.

5.8.1 Pharmacogenomics information

In both personalized medical care and the pharmacogenetics field, databases are of prime importance. The main goal of the field of pharmacogenomics is to find biomarkers that can predict how toxic a drug will be and how the body will react to it. This will make patient treatment more personalized and effective.

Some of the pharmacogenomics Web Resources are:

1. PharmGKB (<https://www.pharmgkb.org/>)
2. CPIC (<https://cpicpgx.org/>)
3. DrugBank (<https://go.drugbank.com/>)
4. SCAN (<http://www.scandb.org/>)
5. PACdb (<http://www.pacdb.org/>)
6. Human Cytochrome P450 Allele Nomenclature database transitioned to PharmVar
7. Cytochrome P450 Drug Interaction Table (<https://druginteractions.medicine.iu.edu/MainTable.aspx>)
8. FDA's pharmacogenetic website (<https://www.fda.gov/drugs/scienceresearch/researchareas/pharmacogenetics/ucm083378.html>)

5.8.1.1 Pharmacogenomics knowledgebase (PharmGKB)

The Pharmacogenomics Knowledgebase (Pharm GKB) contains genetic variants, explication, pharmacological routes, and drug action linkages. The effort is led by Stanford University and supported by the NIGMS (National Institutes of General Medical Sciences). PharmGKB has evolved into a crucial infrastructure for direct treatment over the last 20 years. Pharm GKB seeks to assist researchers in better understanding how genetic differences in people might influence medication responses. Pharm GKB seeks to assist researchers in better understanding how genetic differences in people might influence medication responses. PharmGKB assesses measurement recommendations by releasing dual genetic quotation guidelines and tying them to basic evidence on genetic and related medicines. A drug's name, generic name, short phrase, full description, including label quotes, and, in

the case of US labels, whether the label is listed in the FDA Table for Pharmacogenetic Organizations are all used to define the label. PharmGKB collects data from a variety of sources, but it is up to the user to choose which data to utilize and how to administer the drug. The user is responsible for reviewing and drawing conclusions from all accessible guidelines. The FAIR principles are: available, useable, and active, and PharmGKB is dedicated to following them. These guidelines ensure that PharmGKB data is utilized in a way that may be accessed by others in the present and future. From top genomic and medication services, we give the names, chemical compositions, allele waves, geographical information, and variants (e.g., NCBI, PubChem, Ensemble, HGNC, dbSNP, gnomAD, etc.).

5.8.1.2 *DrugBank*

DrugBank is a comprehensive, publicly accessible digital resource that includes specific drug information, administration, action, and contact information for FDA-approved pharmaceuticals, including testing drugs produced through the FDA-approved method. DrugBank has become one of the most extensively used drug sources in the world due to its extensive, high-quality information. The general public, teachers, pharmacists, medical pharmacists, pharmacists, and the pharmaceutical business all use it regularly. The 2008 release of DrugBank 2.0 covered pharmacological, medical, and chemical information. DrugBank 3.0, which came out in 2010, improves medication-drug interactions, drug delivery information, and pharmacokinetic data. When DrugBank 4.0 came out in 2014, it had a lot of QSAR (work-related activity size) data, information about how drugs work, and ADMET statistics.

DrugBank's purpose is to give a comprehensive collection on pharmaceuticals, including-biochemical data, mechanisms, and targets, Pharmacological data. It concentrates on responses of drug related to changes in the gene of humans or particular polymorphisms.

DrugBank 5.0 has introduced a number of new data types. This contains information about the key drugs' effects on metabolite levels (pharmacometabolomics), genetic levels (pharmacotranscriptomics), and protein synthesis levels (pharmacoproteomics).

Over the years, we've collected and evaluated 27,572 peer-reviewed papers. The DrugBank blocking team has deleted, checked, and installed data from those sources that meet the entrance requirements. Many researchers have used this type of information to test new medication properties or recover old ones. The number of phase I, II, and III research pharmaceuticals in PathWhiz, a JavaScript-based picture rendering application, is used to display all of these ways. The DrugBank prevention team is also stepping up its efforts to collect more information on drugs, drug usage, and drug trafficking. Understanding drug pharmacokinetics, availability, and ADMET symptoms requires this knowledge. Drug linkages with Drug-Bank have been one of the most significant changes or improvements to current Drug-Bank 5.0 data. Details about drug interactions are critical for patients, doctors, and pharmacists, particularly when given to the elderly, and older patients.

5.8.1.3 CPIC

CPIC was formed in 2009 by collaborating with the pharmacogenomics research network (PRN) and the Pharm GKB. The goal of the CPIC is to offer specific gene/drug clinical practice recommendations to help integrate pharmacogenetic research into clinical practice. The CPIC gathers scientific information at all levels, from biological findings into clinical trials, then analyses and incorporates it into the standards. These CPIC recommendations will assist physicians in recognizing how a genetic test can be utilized to enhance medication therapy rather than outlining the reasons for testing. The CPIC currently comprises 174 drug/gene combinations from all CPIC levels, containing 63 genes and 132 therapeutics. Each CPIC recommendation has been reviewed by a group of experts and is kept up-to-date and available to the public on the CPIC website.

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